A CLASSICAL GENERATOR IS A STRONG GENERATOR IN THE PRESENCE OF A STRONG GENERATOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11292-11309, following the statement at 11285-11290.

Editorial note. No omissions: the inclusion chain is displayed in full.

Allowed standard prerequisites (editorial).

- Smallest subcategory containing an object. lemma-find-smallest-containing-E (carded L26204, cited at ~11296). E' lands in some <E>_m.
- Explicit description of the filtration steps. lemma-generated-by-E-explicit (same chapter, cited at ~11305). Index bookkeeping for the star powers.

Source reference. Original lemma 0ATG: sources/stacks-derived/pinned-original-source.tex, source lines 11192–11192. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FX9: sources/stacks-derived/pinned-original-source.tex, source lines 11166–11166. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category which has a strong generator. Let E be an object of $\mathcal{D}$. If E is a classical generator of $\mathcal{D}$, then E is a strong generator.*

Proof. Let E' be an object of $\mathcal{D}$ such that $\mathcal{D} = \langle E' \rangle_n$. Since $\mathcal{D} = \langle E \rangle$ we see that $E' \in \langle E \rangle_m$ for some $m \geq 1$ by Lemma 0ATG. Then $\langle E' \rangle_1 \subset \langle E \rangle_m$ hence

$$\mathcal{D} = \langle E' \rangle_n = smd(\langle E' \rangle_1 \star \ldots \star \langle E' \rangle_1) \subset smd(\langle E \rangle_m \star \ldots \star \langle E \rangle_m) = \langle E \rangle_{nm}$$

as desired. Here we used Lemma 0FX9. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.